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(54) **TOWEL HEATING DEVICE**

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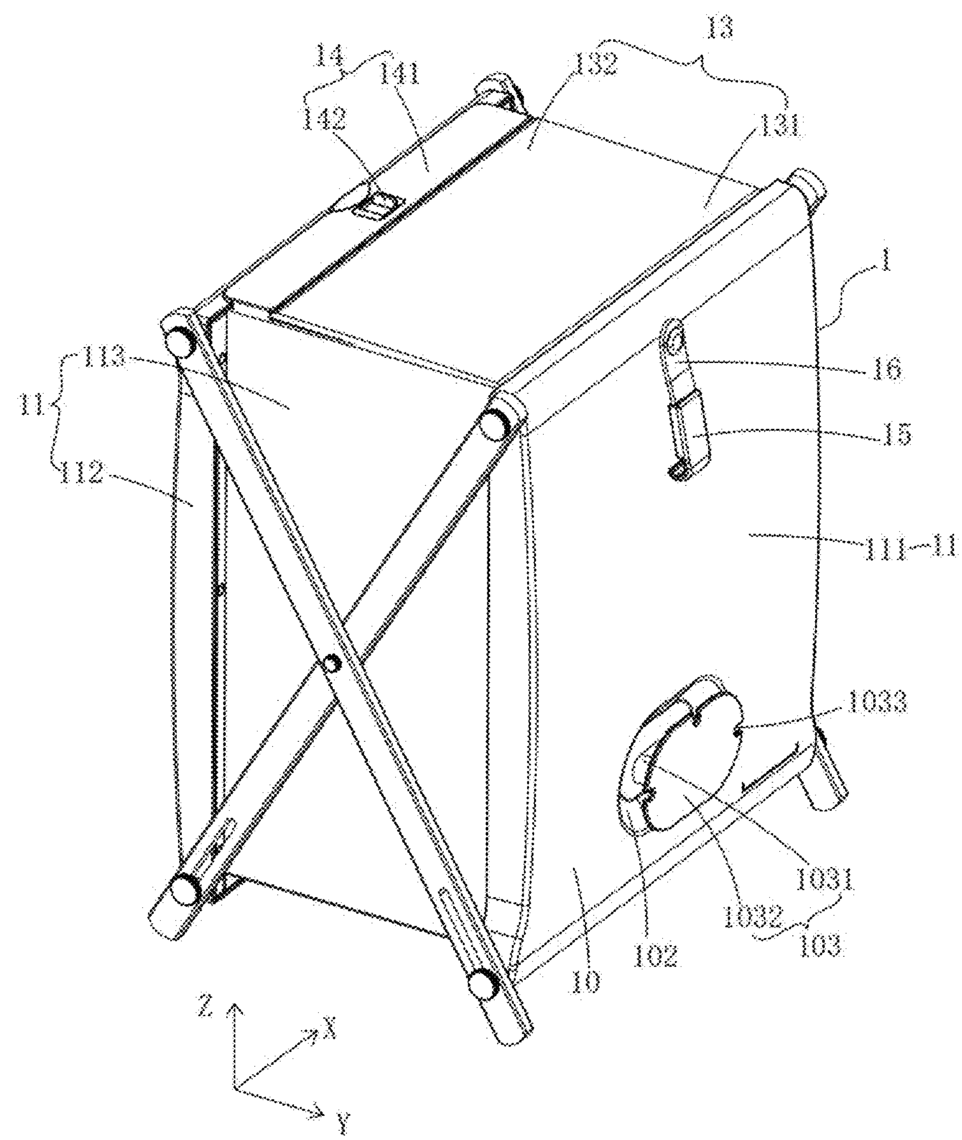
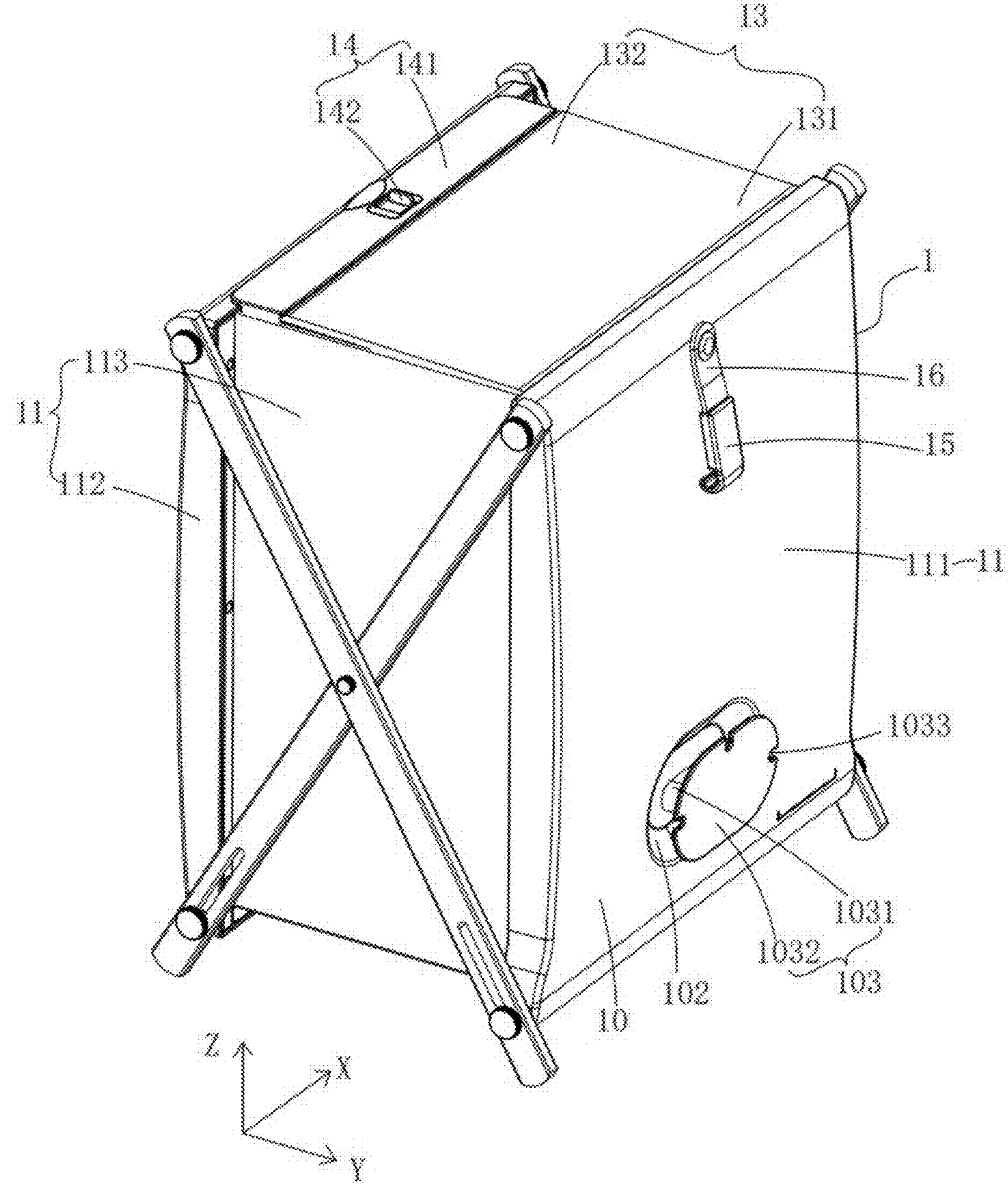
CPC ***A47K 10/06*** (2013.01)

(57) **ABSTRACT**

A towel heating device includes a body portion, at least one heating plate and at least one heat insulation member, the body portion defines a heating chamber; each heating plate is mounted on the body portion; each heat insulation member is mounted in the heating chamber, and the heat insulation member is configured to separate or partially separate the heating plate from the heating chamber.




13
14
141
132
142
131
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113
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112
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1033
1031
1032
103
102
10
Z
X
Y

FIG. 1

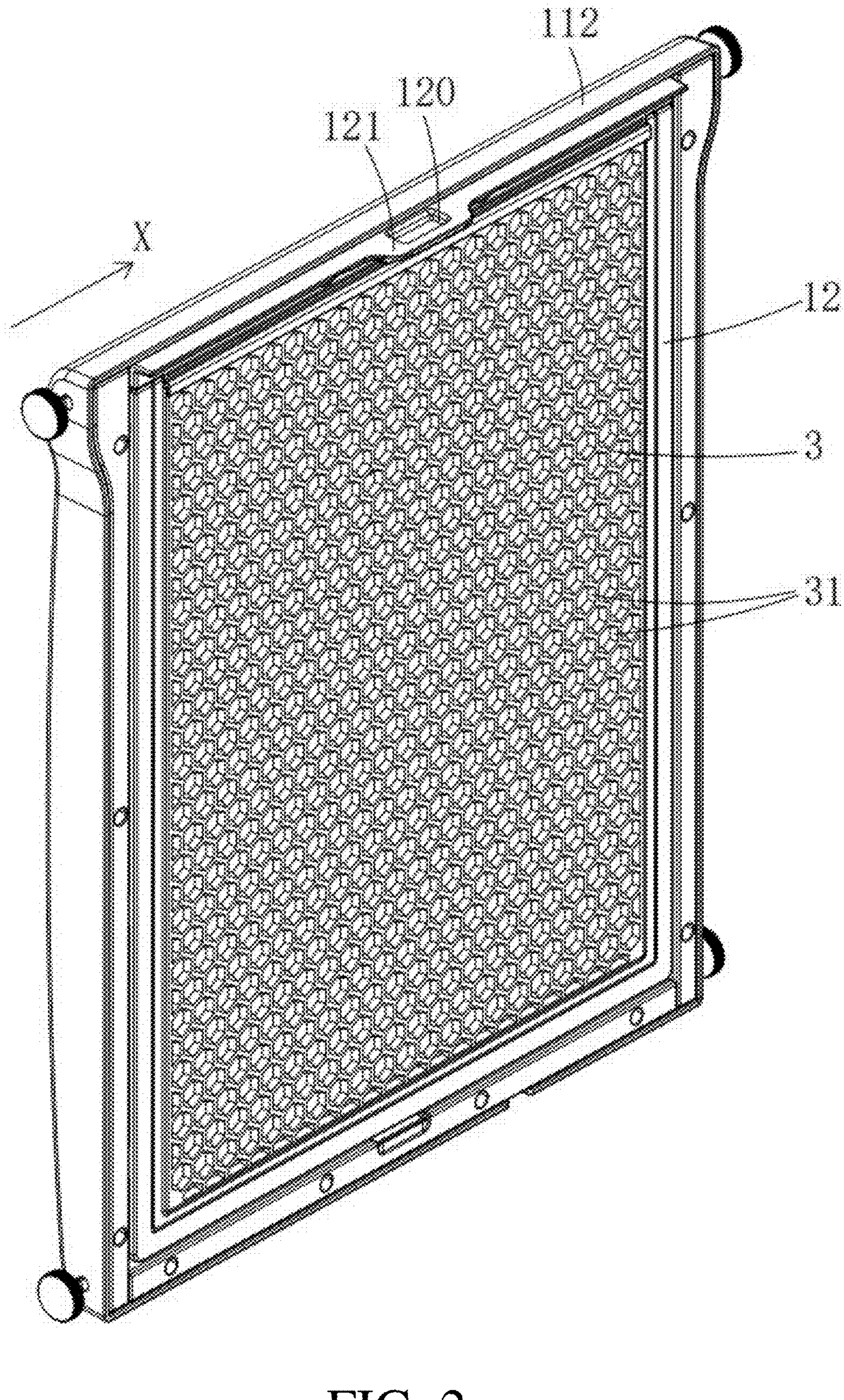

FIG. 2

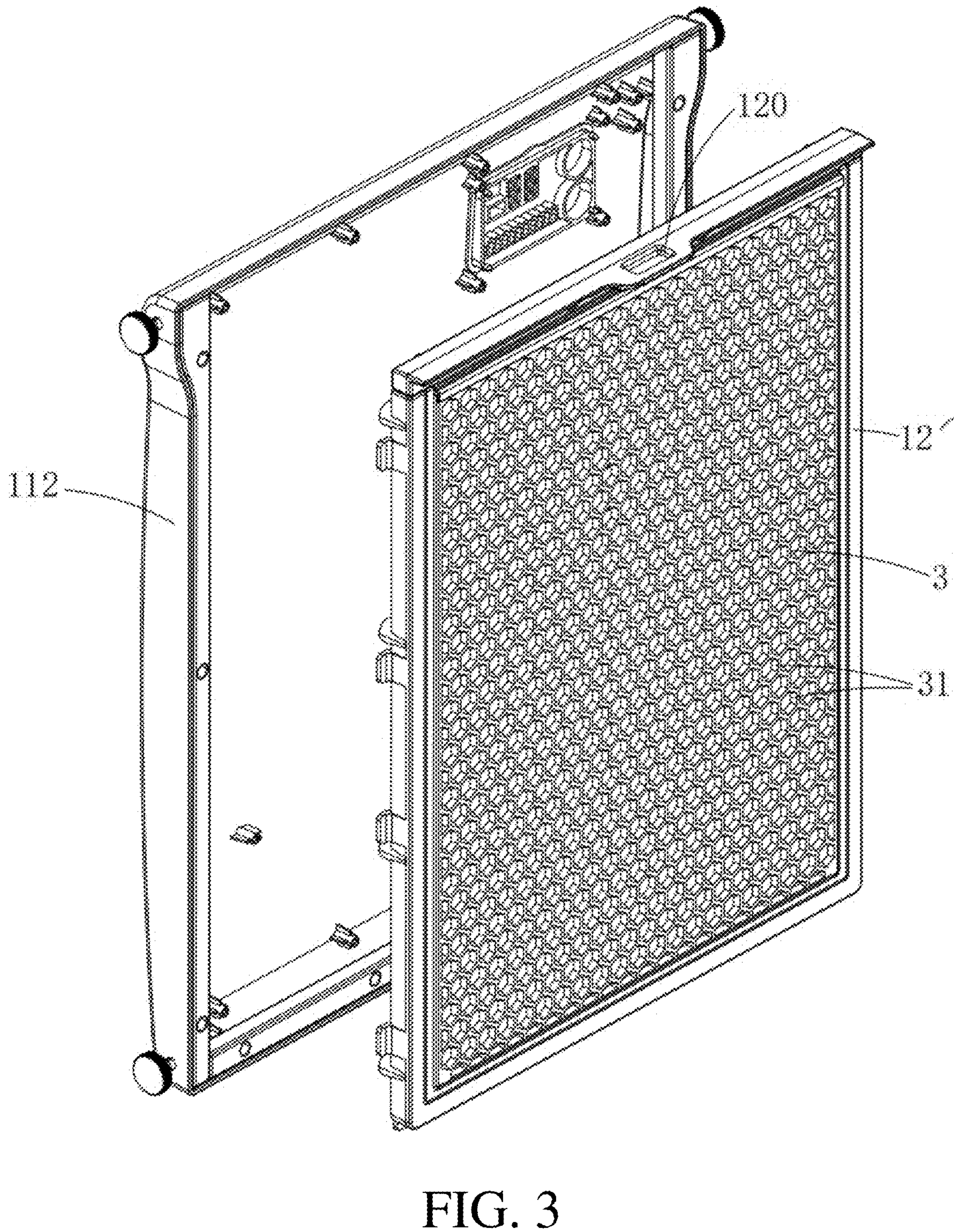

FIG. 3

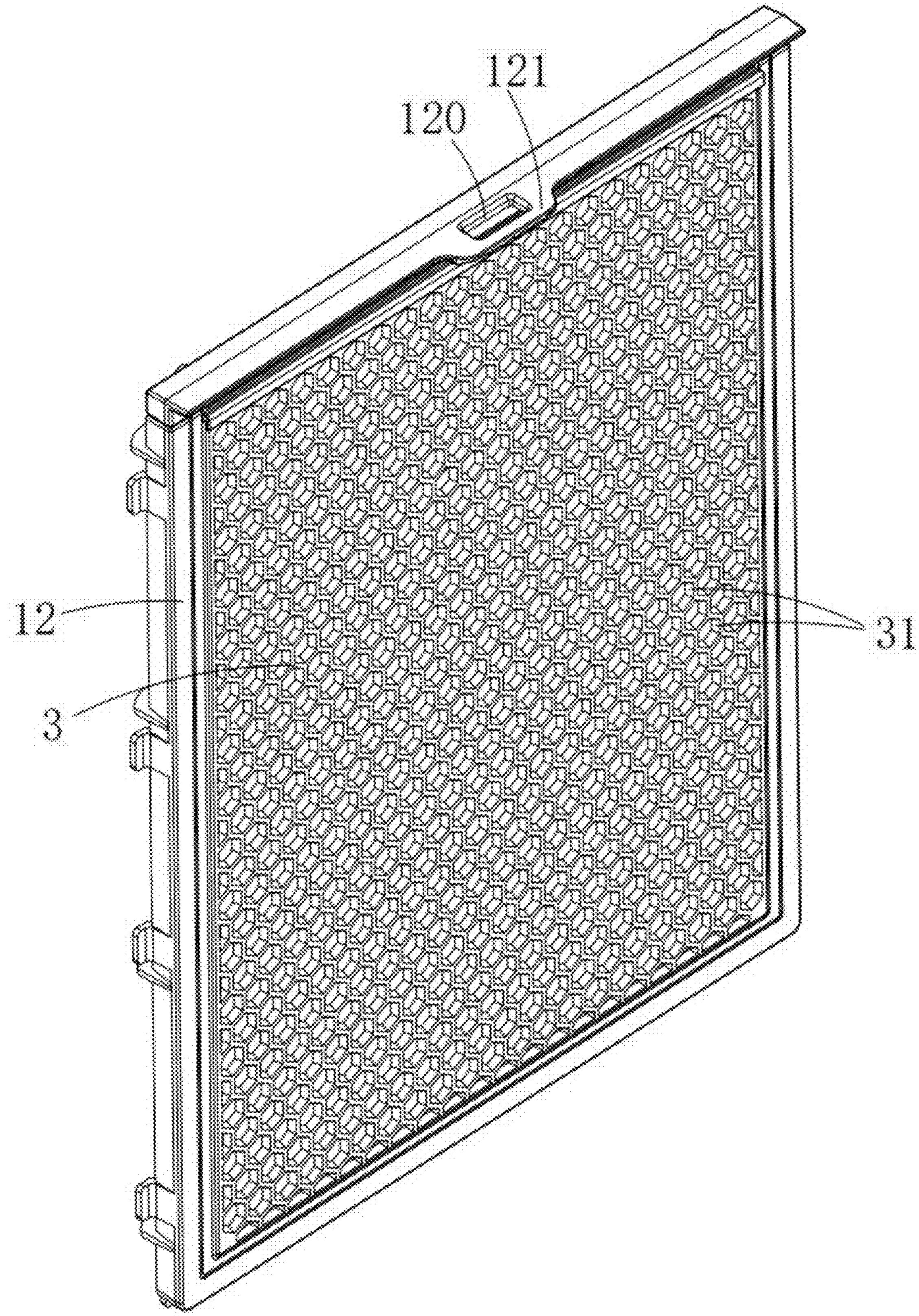

FIG. 4

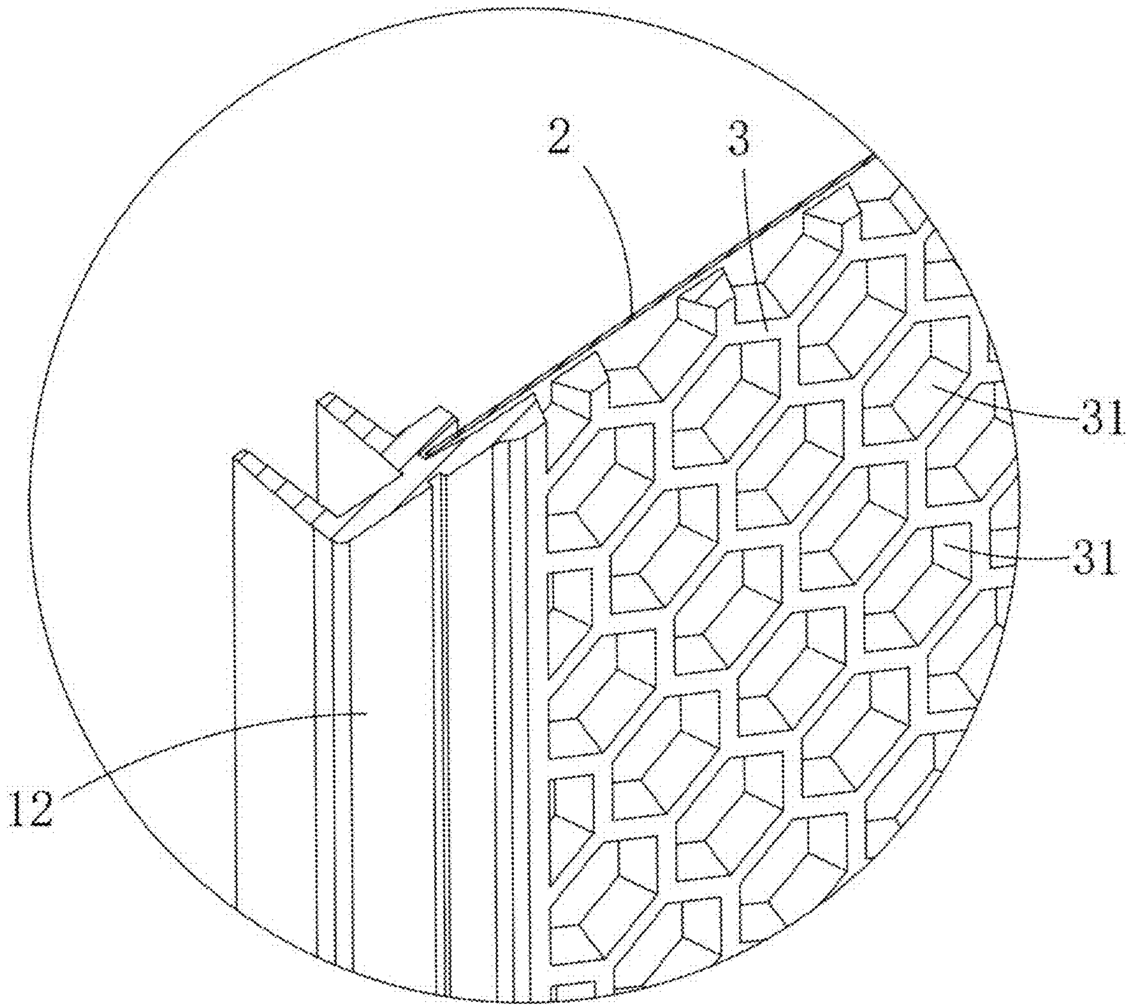

FIG. 5

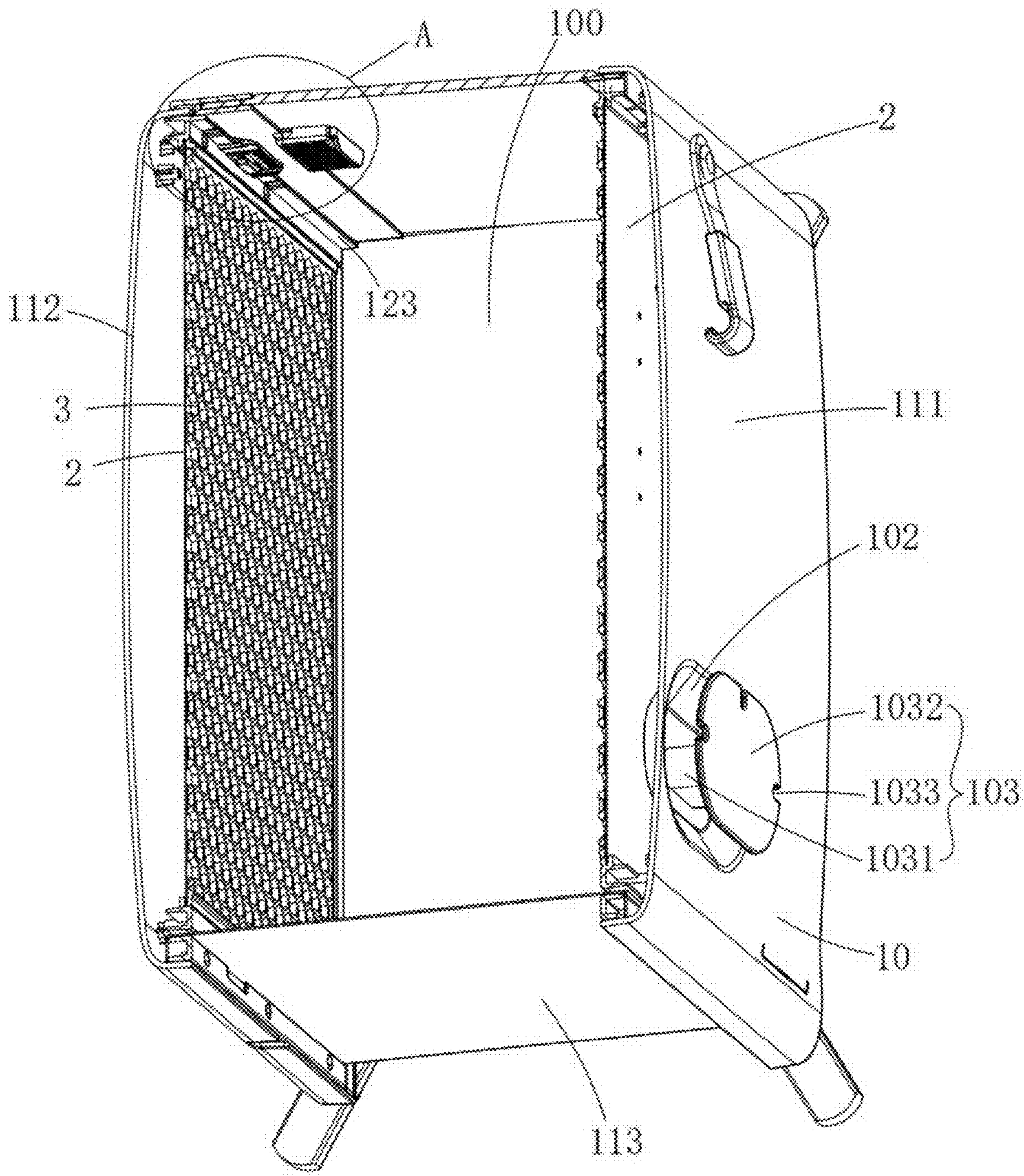

FIG. 6

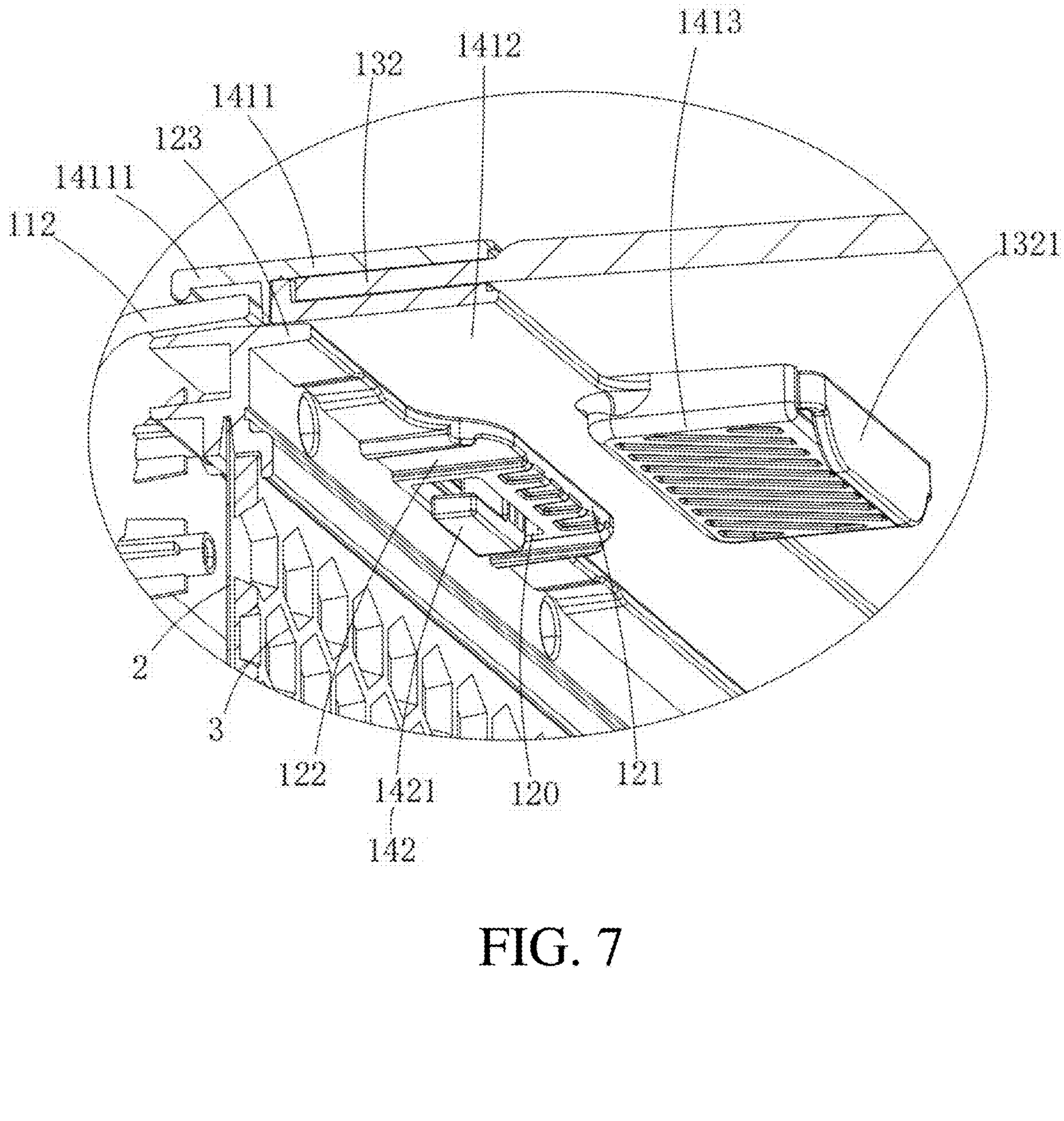

FIG. 7

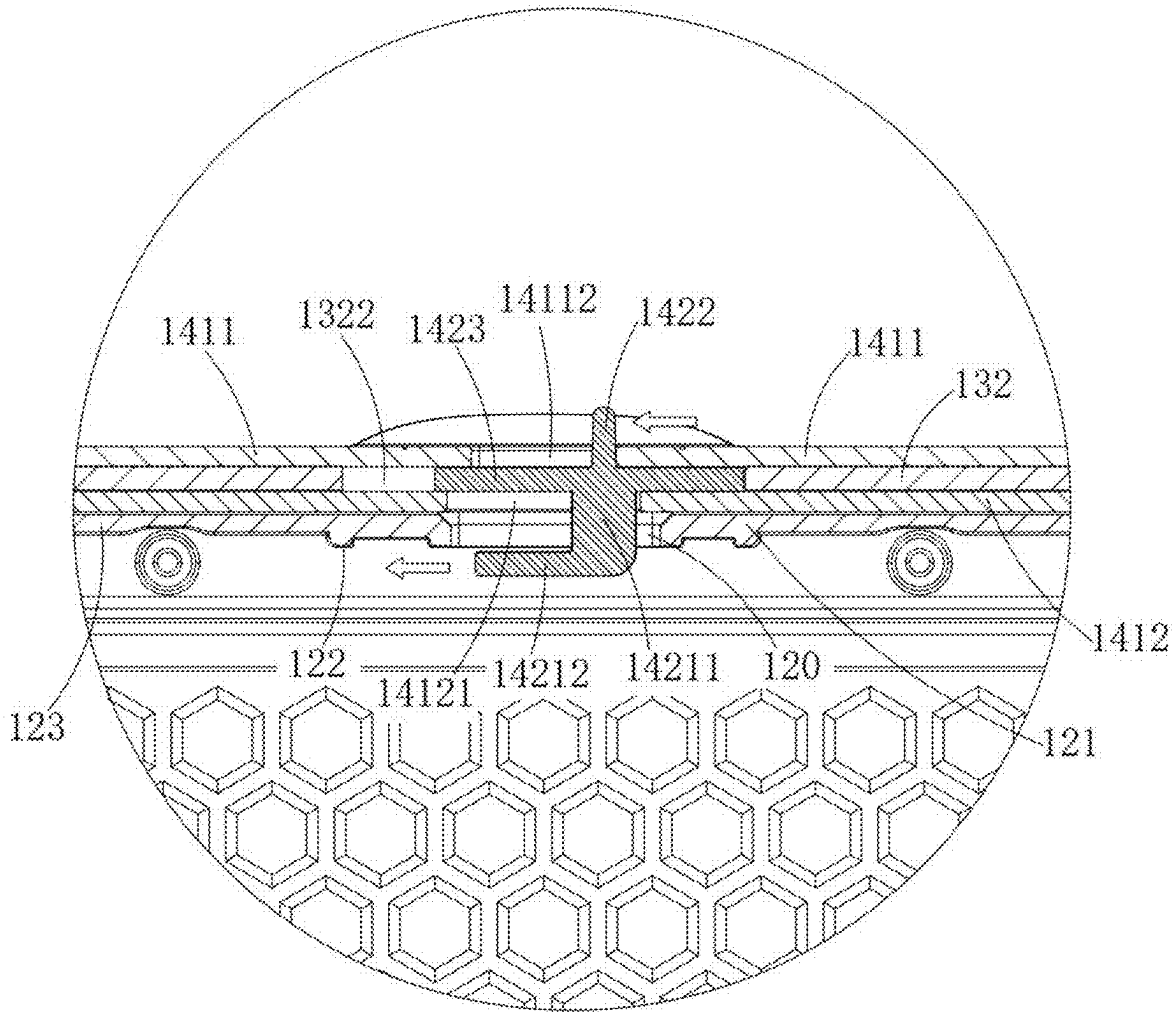
1411
1322
1423
14112
1422
1411
132
1412
123
122
14121
14212
14211
120
121

FIG. 8

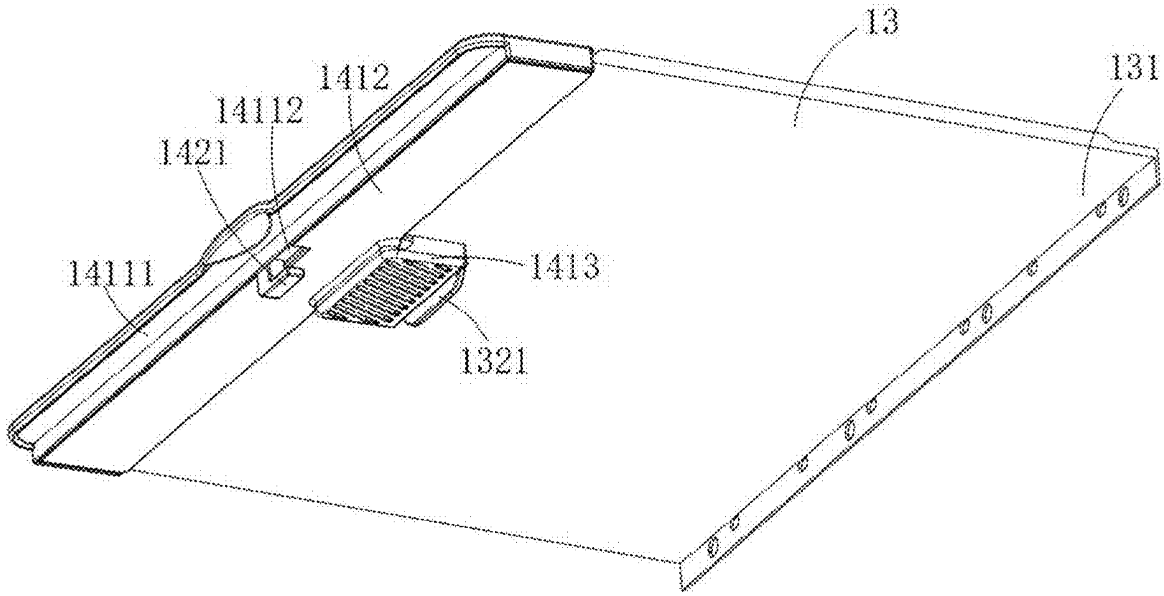
13
131
1412
14112
1421
14111
1413
1321

FIG. 9

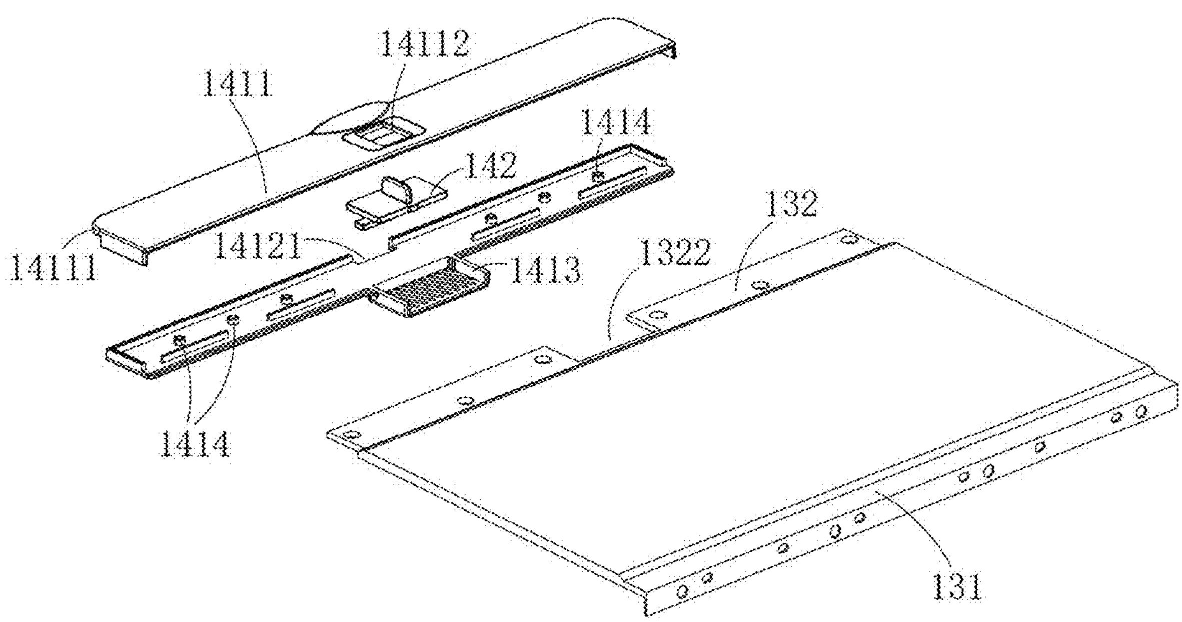
14112
1411
1414
142
132
14111
14121
1413
1322
1414
131

FIG. 10

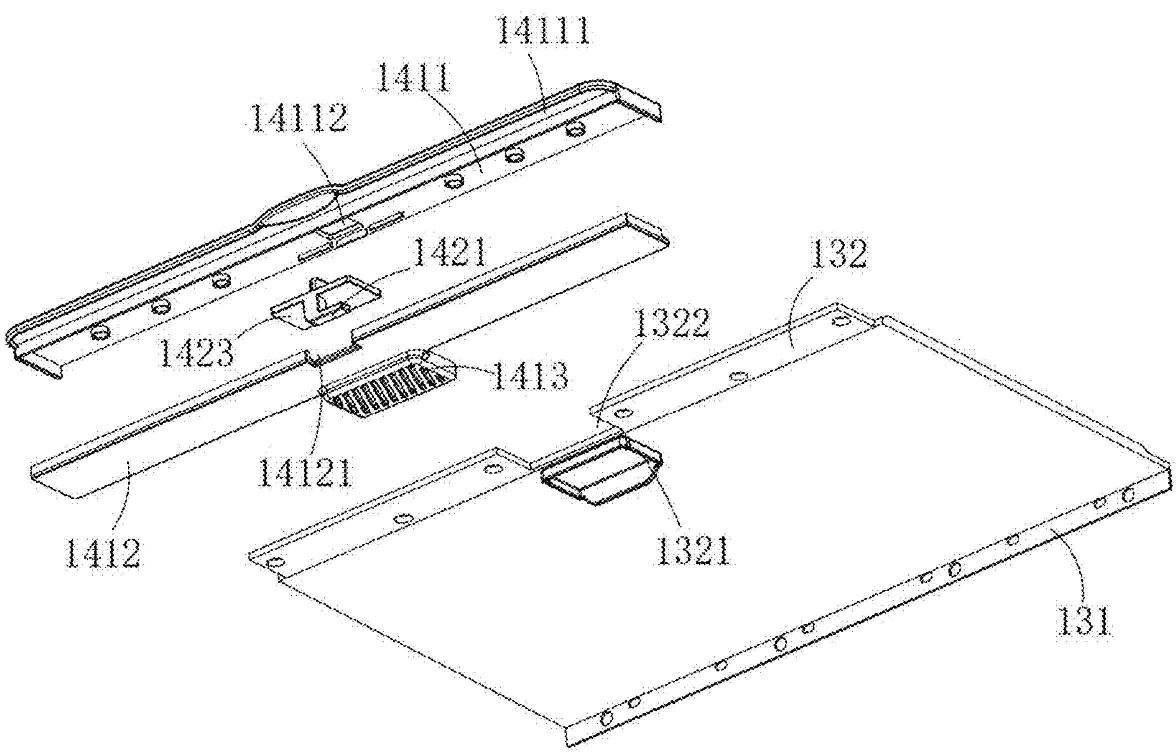

FIG. 11

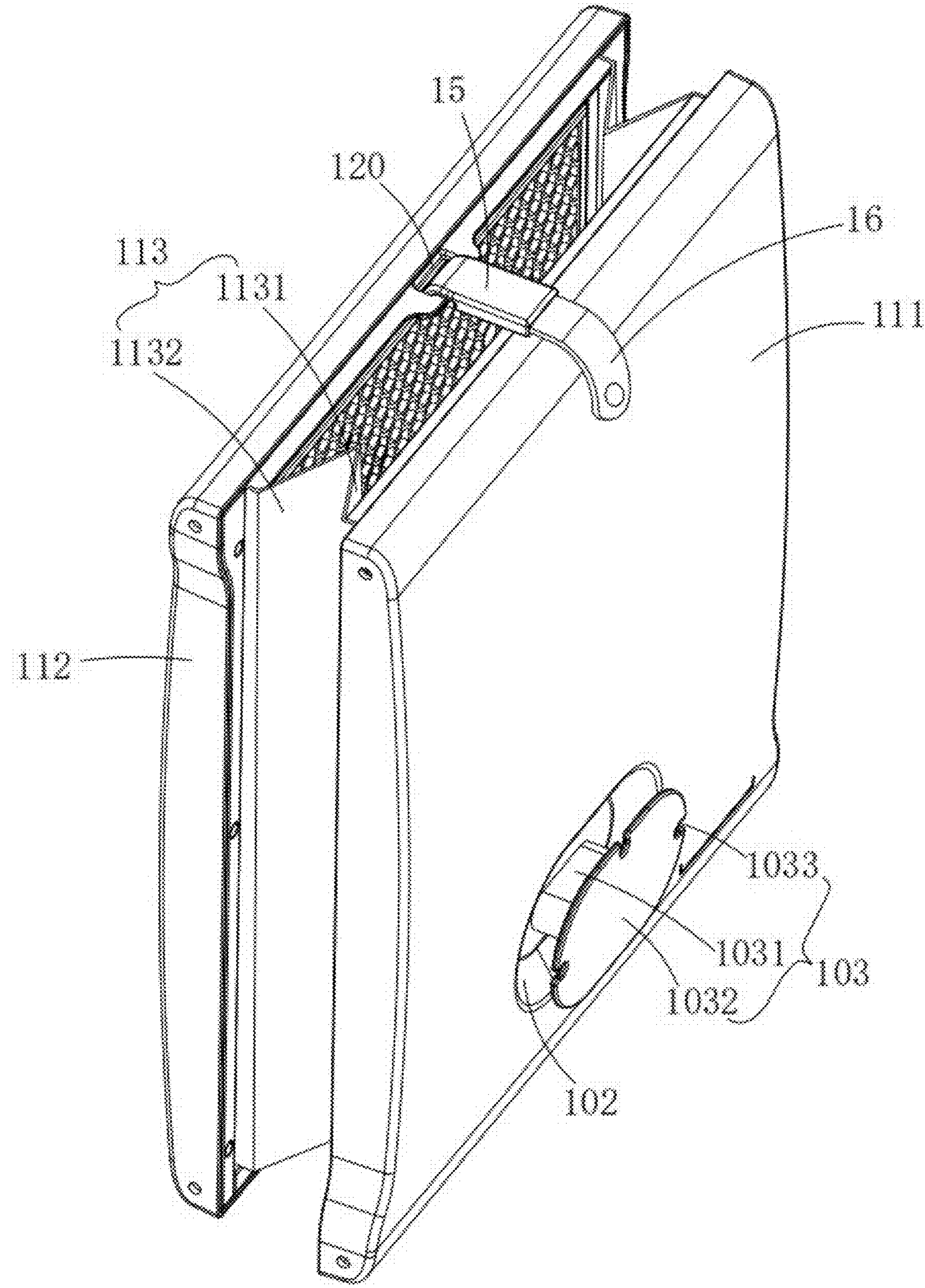

FIG. 12

TOWEL HEATING DEVICE

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation-in-part application of the U.S. patent application Ser. No. 18/763,632 filed on Jul. 3, 2024; U.S. patent application Ser. No. 18/763,734 filed on Jul. 3, 2024, which claims the priority of: Chinese patent application No. CN202410260988.9, filed on Mar. 7, 2024; Chinese patent application No. CN202420441375.0, filed on Mar. 7, 2024; Chinese patent application No. CN202520656668.5, filed on Apr. 8, 2025.

FIELD

[0002] The present disclosure relates to the field of heating device, and particularly to a towel heating device.

BACKGROUND

[0003] The towel heating device has a heating chamber, which is heated by heating wires attached to a metal heating plate. However, in the related art, the metal heating plate of the towel heating device is directly exposed inside the heating chamber, If a user inserts limbs into the heating chamber, they may get burned due to the direct contact with the metal heating plate.

SUMMARY

[0004] A towel heating device includes a body portion, at least one heating plate and at least one heat insulation member, the body portion defines a heating chamber; each heating plate is mounted on the body portion; each heat insulation member is mounted in the heating chamber, and the heat insulation member is configured to separate or partially separate the heating plate from the heating chamber.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] To provide a clearer illustration of the technical solutions in the embodiments of the present disclosure or in the prior art, a brief introduction will be given to the drawings used in the description of the embodiments or the prior art. It is obvious that the drawings described below are merely some embodiments of the present disclosure, and for those skilled in the art, other drawings can be obtained based on these drawings without creative efforts.

[0006] FIG. **1** is a structural schematic view of a towel heating device according to an embodiment of the present disclosure.

[0007] FIG. **2** is a schematic view showing the positional relationship between the second shell, the shell, and heat insulation member according to an embodiment.

[0008] FIG. **3** is an exploded view of the second shell and the installation frame shown in FIG. **2**.

[0009] FIG. **4** is a structural schematic view of the shell and the heat insulation member according to an embodiment.

[0010] FIG. **5** is a horizontal cross-sectional schematic view of the heat insulation and the heating plate according to an embodiment.

[0011] FIG. **6** is a vertical cross-sectional schematic view of the towel heating device.

[0012] FIG. **7** is an enlarged view of a portion A shown in FIG. **6**.

[0013] FIG. **8** is a cross-sectional schematic view of a locking structure according to an embodiment.

[0014] FIG. **9** is a structural view of a cover and the locking structure according to an embodiment.

[0015] FIG. **10** is a first exploded assembly view of the cover and the locking structure according to an embodiment.

[0016] FIG. **11** is a second exploded assembly view of the cover and the locking structure according to an embodiment.

[0017] FIG. **12** is a schematic view of a partial structure in a folded state, showing the hook engaged in the snap slot.

REFERENCE NUMERALS IN THE DRAWINGS

[0018] **1**, body portion; **10**, outer wall of the body portion; **11**, shell; **12**, installation frame; **13**, cover; **14**, locking structure; **15**, hook; **16**, rope; **100**, heating chamber; **101**, opening; **102**, recessed cavity; **103**, wire winding structure; **111**, first shell; **112**, second shell; **113**, foldable member; **120**, snap slot; **121**, connecting protrusion; **122**, bottom surface; **123**, abutment platform; **131**, first end; **132**, second end; **1322**, avoidance groove; **141**, connecting base; **142**, sliding hook; **1031**, winding rod; **1032**, baffle; **1033**, fixation notches; **1321**, engagement block; **1411**, upper base; **1412**, lower base; **1413**, engagement base; **1414**, fixing posts; **1421**, hook-shaped bottom; **1422**, push-pull portion; **1423**, sliding portion; **14111**, lapping plate; **14112**, first sliding slot; **14121**, second sliding slot; **14211**, vertical block; **14212**, stopping block; **1131**, first sheet portion; **1132**, second sheet portion; **2**, heating plate; **3**, heat insulation member; **31**, holes.

DETAILED DESCRIPTION

[0019] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the art to which this disclosure belongs. The terms used in the description of the application herein are intended for describing particular embodiments only and are not intended to limit the present disclosure. In the description, claims, and the above drawings of the present disclosure, the terms "comprising" and "having", as well as their variants, are intended to convey a non-exclusive inclusion. The terms "first", "second", etc., as used herein, are intended to distinguish between different objects, rather than to describe a particular order.

[0020] Reference to "embodiments" herein implies that a particular feature, structure, or characteristic described in connection with an embodiment may be included in at least one embodiment of the present disclosure. The appearance of the phrase at various places in the specification does not necessarily refer to the same embodiment, nor is it a separate or alternative embodiment that is mutually exclusive of other embodiments. One skilled in the art would explicitly and implicitly understand that the embodiments described herein can be combined with other embodiments.

[0021] In order to enable those skilled in the art to better understand the technical solutions of the present disclosure, the technical solutions in the embodiments of the present disclosure will be clearly and completely described below with reference to the accompanying drawings.

[0022] As shown in FIGS. **1** to **12**, the present disclosure provides a towel heating device including a body portion **1**, at least one heating plate **2** and at least one heat insulation member **3**.

[0023] In an embodiment, a heating chamber **100** is defined inside the body portion **1**, a heating plate **2** is mounted on the body portion **1**, and the heating plate **2** is made of metal material, such as a galvanized steel plate. A heating wire is mounted on and attached to the heating plate **2**. The heating wire is able to generate heat when connected to a power source, and the heat is conducted to the heating plate **2** to expand a heating area, such that the heating plate **2** can heat the air inside the heating chamber **100** over a large area. A heat insulation member **3** is mounted in the heating chamber **100**, the heat insulation member **3** is configured to separate or partially separate the heating plate **2** from the heating chamber, that is, the heat insulation member **3** is disposed on a side of the heating plate **2** facing the heating chamber **100**. The heating plate **2** may generate a high temperature during operation, which may causing accidental burns if a limb of the user reaches into the heating chamber **100**. However, a direct contact between the limb and the heating plate **2** can be prevented by the heat insulation member **3**, such that the accidental burns can be avoid.

[0024] In an embodiment, as shown in FIG. **1**, FIG. **2**, FIG. **3** and FIG. **6**, the body portion includes a shell **11** and an installation frame **12**, the shell **11** includes a first shell **111**, a second shell **112** and a foldable member **113** connecting the first shell **111** and the second shell **112** with each other. Both the first shell **111** and the second shell **112** are made of rigid materials, in some embodiments, the first shell **111** and the second shell **112** are made of a plastic; and the foldable member **113** is made of a flexible material, in some embodiments, the foldable member **113** is made of a fabric that includes thermal insulation cotton therein. The first shell **111**, the second shell **112** and the foldable member **113** cooperatively enclose to form the heating chamber **100**, the first shell **111** and the second shell **112** are positioned opposite to each other, that is, the heating chamber **100** is formed between the first shell **111** and the second shell **112**. An opening **101** is defined at a top of the heating chamber **100**, allowing towels to be placed into the heating chamber **100** from the top of the heating chamber **100**. When the first shell **111** and the second shell **112** move toward each other, driving the foldable member **113** to be folded, a capacity of the heating chamber **100** is reduced, such that the towel heating device can be stored conveniently. When the first shell **111** and the second shell **112** move away from each other, driving the foldable member **113** to be unfolded, such that the capacity of the heating chamber **100** is expanded to allow towels to be placed into the heating chamber **100** for drying, heat retention, and so on.

[0025] The towel heating device includes at least one installation frame **12**, and the at least one installation frame **12** is mounted on either the first shell **111** or the second shell **112**. Both the heating plate **2** and the heat insulation member **3** are mounted on the mounting frame **12**, such that the heating plate **2** and the heat insulation member **3** are capable of moving with the first shell **111** or the second shell **112**. In addition, the mounting frame **12** is configured to separate the heating plate **2** from the shell **11**, preventing the heat of the heating plate **2** from being directly transferred to the shell **11**, therefore melting of the shell **11** caused by the high temperature of the heating plate **2** can be prevented, and burns when touching the shell **11** can also be avoided. In an embodiment, the installation frame **12** is made of a plastic material that are high-temperature-resistant and flame-retardant.

[0026] In an embodiment, as shown in FIG. **5**, the heat generated by the heating plate **2** needs to be conducted into the heating chamber **100** through the heat insulation member **3**. Therefore, in the present embodiment, the heat insulation member **3** is configured to contact the heating plate **2**, to prevent an excessively large gap between the heat insulation member **3** and the heating plate **2**, which could affect a heat transfer of the heating plate **2**. In addition, the material of the heat insulation member **3** is the same as that of the installation frame **12**, which is the plastic material with high-temperature resistance and flame-retardant properties.

[0027] Regarding the specific structure between the heat insulation member **3** and the installation frame **12**, in an embodiment, as shown in FIG. **5**, the heat insulation member **3** and the installation frame **12** are configured as an integral and one-piece structure. In another embodiment, which is not illustrated in the figures of the present disclosure, the heat insulation member **3** is connected to an inner side of the installation frame **12** through a detachable connection, for example, through an interference fitting connection, a snap-fit connection or other connections. The difference between the above two embodiments lies only in the installation method and manufacturing cost, which does not affect the function of the heat insulation member **3**.

[0028] In an embodiment, in order to improve the heating efficiency when the heating plate **2** heating the heating chamber **100** through the heat insulation member **3**, a plurality of holes **31** are defined in the heat insulation member **3**, allowing the heating plate **2** to directly heat the air inside the heating chamber **100**.

[0029] The heat insulation member **3** is arranged to fully cover an inner side of the installation frame **12**, the plurality of holes **31** on the heat insulation member **3** are arranged in a honeycomb layout, allowing a diameter of each of the plurality of holes **31** to be configured smaller without affecting the heat insulation of the heat insulation member **3**. In some embodiments, the heat insulation is entirely made of silicone material, and the heat insulation member **3** does not define any hole **31**, that is, the heat insulation member **3** is an integrity board, completely isolating the heating plate **2** from the heating chamber **100**. The heat generated by the heating plate **2** can only be transferred to the heating chamber **100** through the heat insulation member **3**. The embodiment described above is not depicted in the provided figures.

[0030] In order to improve the efficiency of the heating chamber **100** being heated, in some embodiments, the towel heating device includes two heating plates **2**. To correspond with the two heating plates **2**, two heating insulation boards **3** and two installation frame **12** are provided in the present disclosure, therefore, the heating chamber **100** is formed between the two heating plates **2**, with the two heating plates **2** being positioned oppositely to each other.

[0031] To prevent children from accidentally inserting their limbs into the heating chamber **100**, as shown in FIG. **1**, FIG. **6** and FIG. **7**, in an embodiment, the body portion **1** includes cover **13** and locking structure **14**, a first end **131** of the cover **13** is mounted on the first shell **111**, the locking structure **14** is mounted at a second end **132** of the cover **13**. The locking structure **14** is detachably connected to the second shell **112** or the installation frame **12** inside the second shell **112**, to drive the cover **13** to open or close the opening **101**. The cover **13** may be implemented by users to seal the heating chamber **100**, therefore, on the one hand, a heat loss from the heating chamber **100** is reduced, on the other hand, for children, limbs accidentally entering the heating chamber **100** can be prevented. The cover **13** is made of a flexible material, such as a fabric containing thermal insulation cotton on the inside.

[0032] A snap slot **120** is defined in the second shell **112** or the installation frame **12** of the second shell **112**. In some embodiments, as shown in FIGS. **2** to **4**, and FIG. **7**, a connecting protrusion **121** is arranged on the installation frame **12**, and a snap slot **120** vertically extends through the connecting protrusion **121**, such that, the locking structure **14** is able to extend into the snap slot **120**, to achieve a snap-fit engagement.

[0033] In an embodiment, as shown in FIG. **1** and FIG. **7**, the locking structure **14** includes a connecting base **141** and a sliding hook **142**, the sliding hook **142** is slidably mounted on the connecting base **142**, the connecting base **141** is connected to the second end **132** of the cover **13**, the sliding hook **142** has a hook-shaped bottom **1421**, and the hook-shaped bottom is configured to pass through the snap slot and lap over a bottom surface of the connecting protrusion. The connecting base **141** can be placed on the second shell **112** or abut against the installation frame **12** inside the second shell **112**, allowing the cover **13** to close the opening **101**. The hook-shaped bottom **1421** is located below the connecting base **141** and is configured to pass through the snap slot **120**, in this way, moving the sliding hook **142** can drive the hook-shaped bottom **1421** to move beneath the connecting protrusion **121**. When an upward pulling force is applied to the connecting base **141**, the hook-shaped bottom **1421** abuts against the bottom surface **122** of the connecting protrusion **121**, such that, detachment of the connecting base **141** from the second shell **112** is prevented. Therefore, the locking structure **14** can be used to maintain the cover **13** in a locked state.

[0034] Regarding the connection structure between the connecting base **141** and the cover **13**, as shown in FIG. **7**, the connecting base **141** may be configured to clamp the second end **132** of the cover **13**. In an embodiment, the connecting base **141** includes an upper base **1411** and a lower base **1412**. The upper base **1411** and the lower base **1412** cooperatively clamp the second end **132** of the cover **13**, enabling the upper base **1411** to be placed on the second shell **112**, with the lower base **1412** lapping over on the second shell **112** and the installation frame **12** inside the second shell **112**.

[0035] In an embodiment, as shown in FIG. **6** to FIG. **11**, an abutment platform **123** is arranged on an upper side of the installation frame **12**, protruding toward the heating chamber **100** relative to the second shell **112**. The connecting protrusion **121** is formed on the abutment platform **123**, the connecting protrusion **121** and the abutment platform **123** are configured as an integral structure. The upper base **1411** is arranged with a lapping plate **14111** protruding in a direction away from the cover **13**. When the cover **13** closes the opening **101**, the lapping plate **14111** abuts against the second shell **112**, and the lower base **1412** abuts against the abutment platform **123**.

[0036] The sliding hook **142** includes a push-pull portion **1422**, a sliding portion **1423**, and the aforementioned hook-shaped bottom **1421**. The push-pull portion **1422** is formed on an upper side surface of the sliding portion **1423**, and the hook-shaped bottom **1421** is formed on a lower surface of the sliding portion **1423**. The push-pull portion **1422**, the sliding portion **1423**, and the hook-shaped bottom **1421** are configured as an integral structure. The sliding portion **1423** is clamped between the upper base **1411** and the lower base **1412**. Correspondingly, an avoidance groove **1322** is defined at the second end **132** of the cover **13** to accommodate the sliding portion **1423**. A first sliding slot **14112** is defined on the upper base **1411**, the first sliding slot **14112** extends in a length direction X of the towel heating device. The push-pull portion **1422** passes through the first sliding slot **14112** and at least a part of the push-pull portion **1422** is exposed to the outside of the connecting base **141**. The push-pull portion **1422** is able to slide within the first sliding slot **14112**, allowing a user to grip or pinch the push-pull portion **1422** to drive the sliding portion **1423** to slide. A second sliding slot **14121** is defined on the lower base **1412**, and the second sliding slot **14121** also extends in the length direction X of the towel heating device. The hook-shaped bottom **1421** passes through the second sliding slot **14121** and is able to slide within the second sliding slot **14121**. As shown in FIG. **1**, the direction X represents a length direction of the towel heating device, a direction Y represents a width direction of the towel heating device, and a direction Z represents a height direction of the towel heating device.

[0037] The hook-shaped bottom **1421** includes a vertical block **14211** and a stopping block **14212** formed on the vertical block **14211**. The vertical block **14211** sequentially passes through the second sliding slot **14121** and the snap slot **120**, the stopping block **14212** is positioned below the connecting protrusion **121** and extends along the length direction X of the towel heating device. When the lower base **1412** laps over the abutment platform **123**, the entire hook-shaped bottom **1421** passes through the snap slot **120**, enabling the stopping block **14212** to be positioned beneath the connecting protrusion **121**. When an external force is applied to the push-pull portion **1422**, causing the hook-shaped bottom **1421** to slide, the stopping block **14212** moves laterally, enabling at least a part of a vertical projection of the stopping block **14212** to overlap with a vertical projection of the connecting protrusion **121**. At this point, if the connecting base **141** is lifted upward, the stopping block **14212** can abut against the bottom surface **122** of the connecting protrusion **121** to prevent a further upward movement of the connecting protrusion **121**.

[0038] In addition, at least one of the upper base **1411** and the lower base **1412** is arranged with a plurality of fixing posts **1414**. The plurality of fixing posts **1414** can penetrate through the second end **132** of the cover **13** to enhance the connection strength between the cover **13** and the connecting base **141**.

[0039] In an embodiment, an engagement base **1413** is arranged on the lower base **1412**, and an engagement block **1321** is mounted at a lower surface of the cover **13**. When the second end **132** of the cover **13** is clamped between the upper base **1411** and the lower base **1412**, the engagement block **1321** also engages with the engagement base **1413**, therefore, the connection strength between the second end **132** of the cover **13** and the connecting base **141** can be enhanced and maintained.

[0040] In an embodiment, as shown in FIG. **1**, FIG. **7**, and FIG. **12**, the body portion **1** further includes a hook **15**, which is movably mounted on the first shell **111**. To be more precise, the hook **15** is mounted on the first shell **111** through a section of a rope **16**. The hook **15** has a hook-shaped front end, when the locking structure **14** disengages from the snap slot **120**, and when the first shell **111** and the second shell **112** are moved toward each other, the hook **15** can be rotated and positioned above the connecting protrusion **121**, and therefore the hook-shaped front end of the hook **15** can engage with the snap slot **120**. A length of the hook **15** is determined based on a storage state of the towel heating container. When the hook **15** is engaged with the snap slot **120**, the first shell **111** and the second shell **112** cannot be moved away from each other, in this way, the storage state of the towel heating container can be maintained reliably.

[0041] About the folding structure of the foldable member **113**, as shown in FIG. **12**, in an embodiment, the foldable member includes a first sheet portion **1131** and a second sheet portion **1132**. The first sheet portion **1131** is connected to the first shell **111**, and the second sheet portion **1132** is connected to the second shell **112**. The first sheet portion **1131** includes an end which is away from the first shell **111**, and the second sheet portion includes an end which is away from the second shell **112**. The end of the first sheet portion **1131** is connected to the end of the second sheet portion **1132**. When the first shell **111** and the second shell **112** move toward each other, the connection between the first sheet portion **1131** and the second sheet portion **1132** moves inward towards the heating chamber **100**, to drive the first sheet portion **1131** and the second sheet portion **1132** to fold toward each other. In an embodiment, the first sheet portion **1131** and the second sheet portion **1132** are configured as an integral structure, the integral structure is constructed from a fabric that encloses two dense cotton pads. Alternatively, the first sheet portion **1131** and the second sheet portion **1132** are constructed from a fabric that encloses two wooden boards, the two wooden boards are positioned to mated with the locations of the first panel **1131** and the second panel **1132**, facilitating the fold operation of the foldable member **113**.

[0042] As shown in FIG. **1**, a wire winding structure **103** is arranged on an outer wall **10** of the body portion **1**. In an embodiment, the wire winding structure **103** is mounted on the first shell **111**.

[0043] A recessed cavity **102** is defined in the outer wall **10** of the first shell **111**, the wire winding structure **103** includes a winding rod **1031** and a baffle **1032** disposed on the winding rod **1031**, the winding rod **1031** protrudes from the recessed cavity **102**, and the baffle **1032** is located on a side of the winding rod **1031** away from the recessed cavity **102**. Therefore, a powder wire of the towel heating container can be wound around the winding rod **1031** and partially accommodated within the recessed cavity **102**. The recessed cavity **102** facilitates reducing a protrusion length of the winding rod **1031** from a surface of the first shell **111**.

[0044] In addition, a plurality of fixation notches **1033** are defined on the baffle **1032**. A plug portion of the power wire can be engaged in any of the fixing notches **1033** to secure the power wire, preventing the power cord from becoming disordered.

[0045] Obviously, the embodiments described above are only a part of the embodiments of the present disclosure, and not all of them. The accompanying drawings give some embodiments of the present disclosure, but do not limit the patentable scope of the disclosure, which may be realized in many different forms. Rather, these embodiments are provided for the purpose of providing a more thorough and comprehensive understanding of the present disclosure. Although the present disclosure has been described in detail with reference to the foregoing embodiments, it is still possible for a person skilled in the art to modify the technical solutions recorded in the foregoing specific embodiments or to make equivalent substitutions for some of the technical features therein. Any equivalent structure made by utilizing the contents of the specification and the accompanying drawings of the present disclosure, directly or indirectly applied in other related technical fields, are all the same within the scope of the patent protection of the present disclosure.

What is claimed is:

1. A towel heating device, comprising:
a body portion, defining a heating chamber;
at least one heating plate, wherein each heating plate is mounted on the body portion;
at least one heat insulation member, wherein each heat insulation member is mounted in the heating chamber, the heat insulation member is configured to separate or partially separate the heating plate from the heating chamber.

2. The towel heating device according to claim **1**, wherein the body portion comprises a shell and an installation frame, the heating chamber is defined in the shell, the installation frame is mounted inside the shell, both the heating plate and the heat insulation member are mounted on the installation frame.

3. The towel heating device according to claim **1**, wherein the heat insulation member is in contact with the heating plate.

4. The towel heating device according to claim **2**, wherein the heat insulation member and the installation frame are configured as an integral and one-piece structure.

5. The towel heating device according to claim **2**, wherein the heat insulation member is detachably mounted in an inner side of the installation frame.

6. The towel heating device according to claim **1**, wherein the heat insulation member defines a plurality of holes.

7. The towel heating device according to claim **6**, wherein the plurality of holes on the heat insulation member are arranged in a honeycomb layout.

8. The towel heating device according to claim **1**, wherein the heat insulation member is made of high-temperature-resistant plastic material.

9. The towel heating device according to claim **1**, wherein the heat insulation member is made of silicone material, and configured to separate the heating plate from the heating chamber.

10. The towel heating device according to claim **1**, wherein the number of at least one heating plate is two, the heating chamber is formed between the two heating plates, and the number of at least one heat insulation member is two, mating with the two heating plates.

11. The towel heating device according to claim **1**, wherein the body portion further comprises a first shell, a second shell, and a foldable member connecting the first shell and the second shell with each other, wherein, the first shell, the second shell and the foldable member cooperatively enclose to form the heating chamber, the first shell and the second shell are configured to move toward or away from each other, driving the foldable member to be folded or unfolded, the heat insulation member is at least mounted on one of the first shell and the second shell.

12. The towel heating device according to claim **11**, wherein the body portion further comprises a cover and a locking structure, an opening is defined on the top of the heating chamber, a first end of the cover is mounted on the first shell, and the locking structure is arranged at a second end of the cover, the locking structure is configured to be detachably connected to the second shell or an installation frame inside the second shell, to drive the cover to open or close the opening.

13. The towel heating device according to claim **12**, wherein a snap slot is defined in the second shell or the installation frame inside the second shell, and the locking structure is configured to extend into and engage with the snap slot.

14. The towel heating device according to claim **13**, wherein a connecting protrusion is arranged on the installation frame, and the snap slot vertically penetrates the connecting protrusion;

wherein the locking structure comprises a connecting base and a sliding hook, the sliding hook is slidably mounted on the connecting base, the connecting base is connected to the second end of the cover, the sliding hook has a hook-shaped bottom, and the hook-shaped bottom is configured to pass through the snap slot and abut against a bottom surface of the connecting protrusion.

15. The towel heating device according to claim **13**, wherein the body portion further comprises a hook movably mounted on the first shell, when the first shell and the second shell move toward each other, and the locking structure is disengaged from the snap slot, the hook is capable of being engaged with the snap slot.

16. The towel heating device according to claim **11**, wherein the foldable member comprises a first sheet portion and a second sheet portion, the first sheet portion is connected with the first shell, the second sheet portion is connected with the second shell, an end of the first sheet portion away from the first shell is connected to an end of the second sheet portion away from the second shell, when the first shell and the second shell move toward each other, a connection between the first sheet portion and the second sheet portion moves toward the heating chamber, to drive the first sheet portion and the second sheet portion to fold toward each other.

17. The towel heating device according to claim **1**, wherein a wire winding structure is arranged on an outer wall of the body portion.

18. The towel heating device according to claim **17**, wherein a recessed cavity is defined in the outer wall of the body portion, and the wire winding structure comprises a winding rod and a baffle disposed on the winding rod, the winding rod extends outward from the recessed cavity, with a portion of the winding rod remaining inside the recessed cavity, and the baffle is located on a side of the winding rod away from the recessed cavity.

19. The towel heating device according to claim **18**, wherein a plurality of fixation notches are defined on the baffle.

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